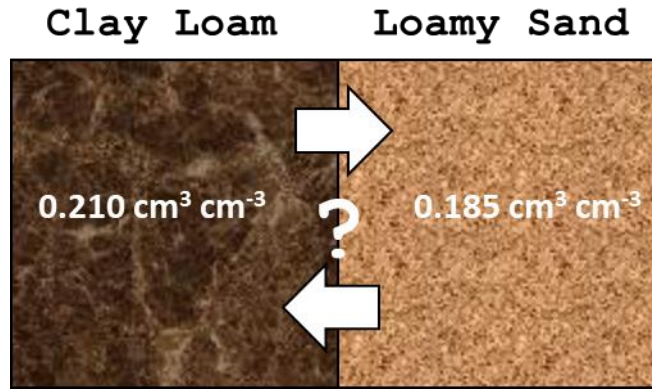


## Practice Questions for the Final Exam

### Questions:

1. Name at least five soil physical properties
2. Name at least five soil physical processes
3. Estimate the drainage rate for a sandy loam soil that has volumetric water content of  $0.250 \text{ cm}^3 \text{ cm}^{-3}$ . Assume a unit gradient of water potential and saturation of  $0.45 \text{ cm}^3 \text{ cm}^{-3}$ .
4. Water moves along \_\_\_\_\_ gradients
5. Heat moves along \_\_\_\_\_ gradients
6. Define what is a “system at equilibrium”.
7. A soil showing steady-state water flow means that is at equilibrium. True/False
8. Define vapor pressure deficit and briefly explain why is important.
9. What is the difference between relative humidity and vapor pressure deficit?
10. What do Fick’s, Darcy’s, and Fourier’s law have in common?
11. Name and define the predominant modes of heat transport in the atmosphere and the soil?
12. Calculate the steady-state evaporation rate from a silt loam soil that has a water table at 2 m depth. Assume that  $10.2 \text{ cm} = 1 \text{ kPa}$
13. Soils having a shallow water table and high evaporation rate have a high risk of developing what environmental process?
14. Name the three conditions required for soil evaporation?
15. Name the process that links the soil water balance with the soil surface energy balance?
16. Name a process that links the energy, water, and carbon cycles?
17. Which are the units of: force, pressure, energy, and power.
18. Depict the soil-plant-atmosphere continuum and label each component with approximate, but realistic, water potential values.
19. Approximate the temperature of the soil surface on April 1. Assume an annual oscillation, annual average soil surface temperature of 13 degrees Celsius, a temperature amplitude at the surface of 15 degrees Celsius, and that the minimum temperature of the year is achieved on January 1.
20. The amplitude of soil temperature as a function of depth decreases/increases exponentially.
21. Define albedo and explain what factors affect it. What is the approximate range for soil?
22. Briefly explain the concept of volumetric heat capacity and soil thermal conductivity.
23. Calculate bulk density, porosity, gravimetric water content, and volumetric water content for a cylindrical soil sample 10.0 cm long and 7.50 cm in diameter, with a wet mass of 685 g and an oven-dry mass of 572 g. Assume a particle density of  $2.65 \text{ g/cm}^3$ . Assume the wet and dry mass does not contain the mass of the ring.
24. Define capillarity. Explain why it occurs. Describe one important effect of capillary rise.

25. Calculate soil water storage for the following soil profile. The A horizon is 32.0 cm thick and has a volumetric water content of  $0.18 \text{ cm}^3 \text{ cm}^{-3}$ . The B horizon is 30.0 cm thick and has a volumetric water content of  $0.17 \text{ cm}^3 \text{ cm}^{-3}$ .
26. Define soil water potential. Define each of its three primary components. What are its units?
27. Draw a figure showing typical soil water retention curves for a coarse-textured soil and a fine-textured soil. Draw both figures in the same chart. Label the axes. Specify the units and choose realistic maximum and minimum values for each axis. Write a short paragraph in which you define the soil water retention curve and describe the differences between the curves for coarse-textured soil versus fine-textured soil.
28. A loamy sand soil with a volumetric water content of  $0.185 \text{ cm}^3 \text{ cm}^{-3}$  is in contact (see sketch) with a clay loam soil that has a volumetric water content of  $0.210 \text{ cm}^3 \text{ cm}^{-3}$ . In what direction does the water move? Assume no solutes.
  - a. Water moves from the Loamy sand to the Clay loam soil
  - b. Water moves from the Clay loam to the Loamy sand soil
  - c. There is no net movement because the soil is already in hydraulic equilibrium



29. Hint: The soils are side-by-side, which means that the gravitational potential is the same for both soils
30. What are the three main stages of soil erosion? For erosion by water, describe the primary processes or factors involved with each of these three stages.
31. Define rainfall interception and list two factors that influence its magnitude.
32. Before a 1.00 inch rainfall event, the soil water content was  $0.15 \text{ m}^3 \text{ m}^{-3}$  in the top 20.0 cm of soil. After the event, the soil water content was  $0.25 \text{ m}^3 \text{ m}^{-3}$  in the top 20.0 cm of soil. Assume there were no losses except interception. What percentage of the rainfall was intercepted?
33. Define rainfall erosivity and specify three factors which influence it.
34. Define soil erodibility and specify three factors which influence it.
35. Use the unit-gradient approach and Campbell's hydraulic conductivity model to:
  - a. estimate the drainage rate for a loam soil with  $K_s = 310 \text{ mm/d}$ ,  $b = 4.5$ ,  $\theta_{sat} = 0.46$ , and  $\theta = 0.40$
  - b. estimate the volumetric water content (an estimate of the field capacity) at which the drainage rate would be 1 mm/day

36. A uniform silt loam soil is initially wetted to  $\theta=0.42 \text{ cm}^3/\text{cm}^3$  to a depth of 1.2 m. The soil has a  $K_s$  value of 160 mm/day,  $\theta_{sat} = 0.44 \text{ cm}^3 \text{ cm}^{-3}$ , and  $b = 4.7$  in the Campbell hydraulic conductivity model.
- What is the value of  $\theta$  when the drainage rate drops to 1 mm/day assuming unit-gradient conditions?
  - How many mm of water have drained from the soil profile when that value of  $\theta$  is reached?
37. A young corn plant in a growth chamber produced 31 g of above-ground biomass with a cumulative transpiration of 4500 g of water.
- What is the water use efficiency (WUE) in  $\text{g kg}^{-1}$ ?
  - For the same growing conditions, would you expect higher or lower WUE for a native C3 plant? Explain why in one sentence.
  - If the vapor pressure deficit during the growth period were increased, would the WUE be higher or lower? Explain why in one sentence.